

2.6 Matrix Multiplication Properties



Core skills you'll master in this section:

- ▶ Consecutive matrix multiplication: several geometric operations are applied to a vector in sequence, taking effect from right to left.
- ▶ The transpose of a matrix product: the order reverses, letting us understand the transformation from a row-vector perspective.
- ▶ Associativity: changing where the parentheses go doesn't change the result, so we can optimize the computation order to reduce the workload.
- ▶ Matrix decomposition: breaking a complex matrix down into basic geometric transformations gives a more intuitive understanding of what it does.
- ▶ Commutativity generally does not hold: even when both products exist, AB and BA can carry different geometric meanings.
- ▶ In special cases, commutativity does hold.
- ▶ Matrix power: repeatedly multiplying a square matrix by itself represents the same linear transformation being applied over and over.

With the geometric perspective on matrix multiplication laid out, understanding the common properties of matrix multiplication becomes much easier.

Consecutive Matrix Multiplication: Sequential Geometric Transformations

Multiplying several matrices in sequence is equivalent to applying these geometric transformations to a geometric object one after another.

Let's take the following three 2×2 matrices as an example

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$$

Matrix $\mathbf{A}$ corresponds to rotation about the origin (rotate); $\mathbf{B}$ corresponds to shearing (shear); $\mathbf{C}$ corresponds to scaling (scale).

Specifically, matrix $\mathbf{A}$ corresponds to a 90° counterclockwise rotation about the origin.

Matrix $\mathbf{B}$ corresponds to shearing along the x_1 horizontal axis. Put simply, shearing is a transformation that tilts a shape along a certain direction, turning originally vertical or horizontal line segments into slanted ones while keeping their parallel relationships unchanged.

Matrix $\mathbf{C}$ shrinks the horizontal axis to $1/2$ and stretches the vertical axis to $2\times$.

If column vectors $\mathbf{x}$ and $\mathbf{y}$ both represent a point in the plane, then the following matrix multiplication

$$A @ B @ C @ x = y$$

represents applying to column vector x scaling (C), then shearing (B), then rotation (A), in that order.

The entire geometric-transformation process is shown in Figure 1.

⚠ Please pay special attention to the order of matrix multiplication $ABCx = y$ — from right to left, that is, $C \rightarrow B \rightarrow A$.

? Please compute by hand the matrix-multiplication results for ABC , CBA , ACB , BAC , and other such permutations.

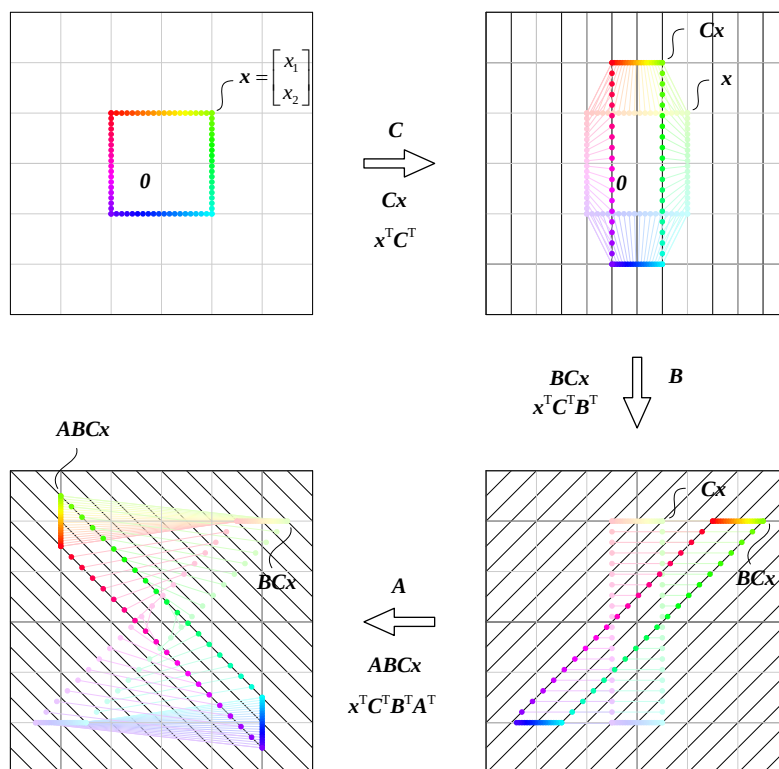
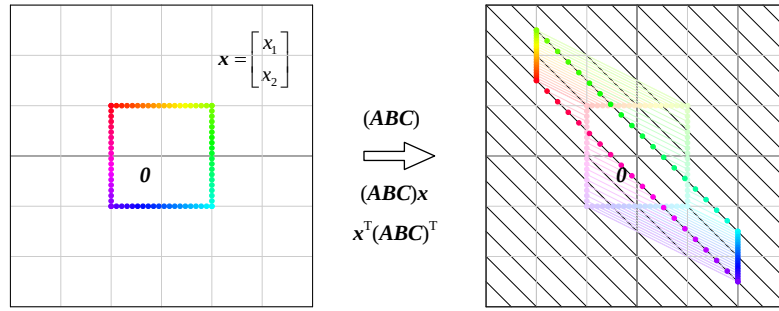


Figure 1. $ABCx = y$'s Step-by-Step Geometric Operations

Of course, we could also compute (ABC) first, and then apply (ABC) as a single whole to x .

Matrix multiplication (ABC) is equivalent to a composite geometric operation — it accomplishes the geometric transformation “in one step”! This is shown specifically in Figure 2.

Figure 2. $(ABC)x = y$'s "One-Step" Geometric Operation

The Transpose of a Matrix Product

The transpose of a matrix product has the following important properties worth paying attention to:

$$\begin{aligned}(AB)^T &= B^T A^T \\ (ABC)^T &= C^T B^T A^T \\ (A_1 A_2 A_3 \cdots A_k)^T &= A_k^T \cdots A_3^T A_2^T A_1^T\end{aligned}$$

This set of properties really doesn't need to be memorized by rote!

Let's take as an example, and transpose this matrix product

$$x^T @ C^T @ B^T @ A^T = y^T$$

Column vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ transposed gives the row vector $x^T = [x_1 \ x_2]$, representing a point on the horizontal plane. The equation above tells us that for the row vector, applying scaling (C), then shearing (B), then rotation (A) in turn, the result is still as shown in Figure 1.

Associativity

In matrix operations, if three matrices A , B , and C are being multiplied together, whether we first multiply A and B , then multiply by C , or first multiply B and C , then left-multiply by A , the final result is the same

$$(AB)C = A(BC)$$

In other words, whether we first compute the product of the first two matrices or first compute the product of the last two matrices, the result is unaffected. This property lets us simplify matrix operations, giving us the flexibility to choose the computation order to improve efficiency or make deriving mathematical formulas easier. At the end of this section, we'll cover how associativity can improve computational efficiency in consecutive matrix multiplication.

⚠ Note that the order of the matrices cannot be changed. That is, matrix multiplication generally does not satisfy commutativity, that is, $AB \neq BA$.

How should we understand ?

Once again from a geometric perspective, (AB) and (BC) are each equivalent to a "local" composite geometric transformation.

Again taking $\mathbf{A}$ as an example, $(\mathbf{AB})$ is equivalent to composing rotation and shearing (with shearing applied first); $(\mathbf{BC})$ is equivalent to composing shearing and scaling (with scaling applied first).

For a planar column vector $\mathbf{x}$, $(\mathbf{AB})\mathbf{C}\mathbf{x}$ is equivalent to first scaling $\mathbf{x}$ by $(\mathbf{C})$, and then applying $(\mathbf{AB})$ — shown specifically in Figure 3.

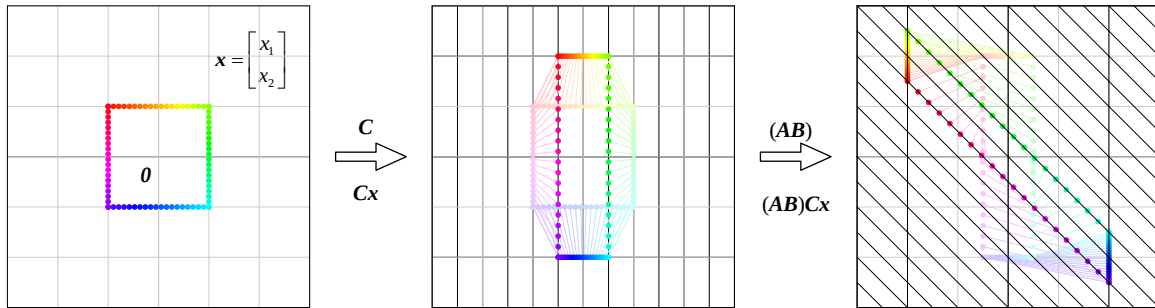


Figure 3. $(\mathbf{AB})\mathbf{C}\mathbf{x} = \mathbf{y}$'s Step-by-Step Geometric Operations

For a planar column vector $\mathbf{x}$, $\mathbf{A}(\mathbf{BC})\mathbf{x}$ is equivalent to applying to $\mathbf{x}$ first $(\mathbf{BC})$, and then rotating $(\mathbf{A})$ — shown specifically in Figure 4.

Comparing Figure 1, Figure 2, Figure 3, and Figure 4 — even though the process involves both expanding and combining terms — we find their results are completely identical.

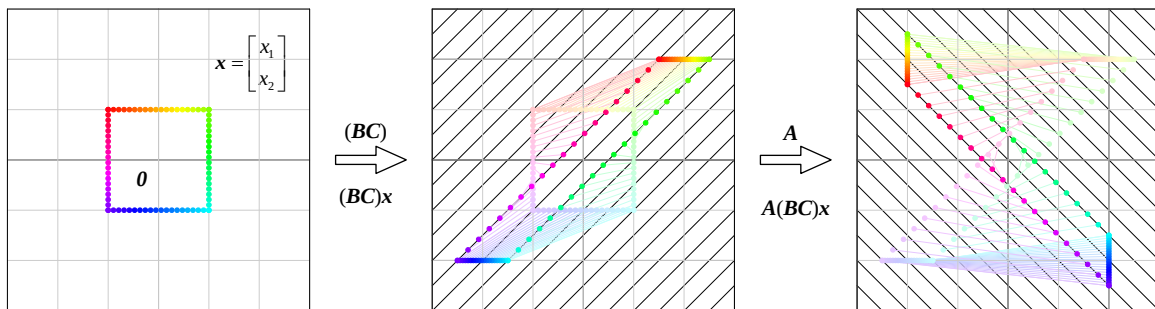


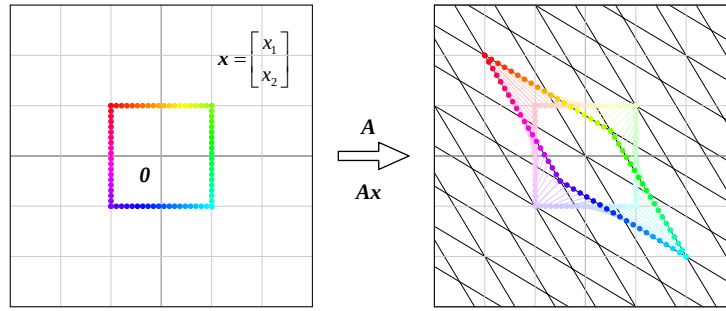
Figure 4. $\mathbf{A}(\mathbf{BC})\mathbf{x} = \mathbf{y}$'s Step-by-Step Geometric Operations

Matrix Decomposition

Given matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

As shown in Figure 5, $\mathbf{A}\mathbf{x} = \mathbf{y}$ is clearly a geometric transformation we're not familiar with.

Figure 5. $Ax = y$'s Geometric Operation

To understand $Ax = y$, we need to use matrix decomposition to break Figure 5 down into geometric transformations we're familiar with.

Put simply, matrix decomposition breaks a matrix down into the product of several matrices. You can think of matrix decomposition as the inverse operation of consecutive matrix multiplication.

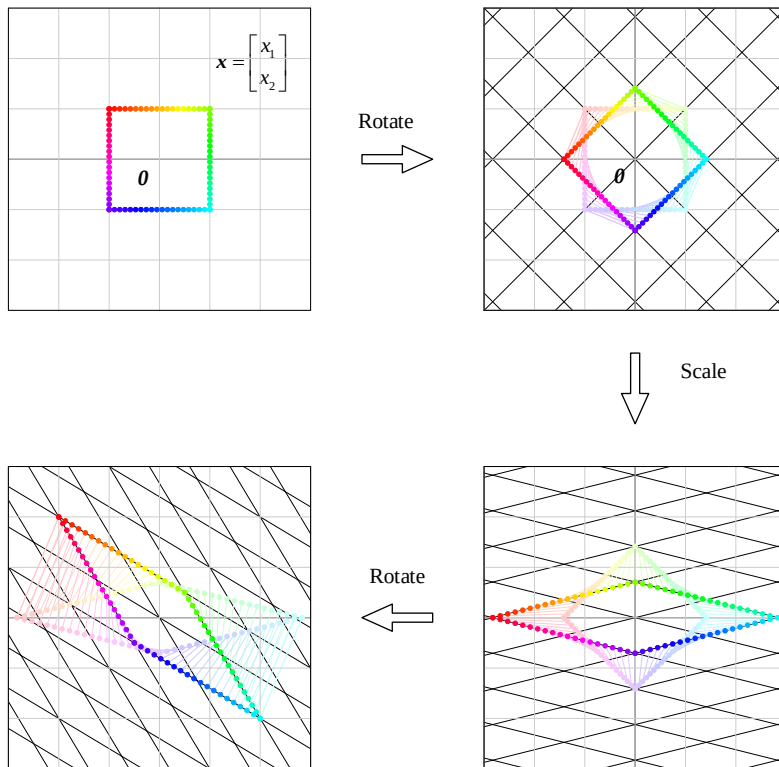
From a geometric perspective, a “complex” geometric operation can be decomposed into several “familiar” geometric operations.

For example, as shown in Figure 6, we can decompose matrix A into “rotation → scaling → rotation”, with the corresponding matrix multiplication being

$$\begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} @ \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

Note that in Figure 6, the two rotations are in exactly opposite directions.

the corresponding decomposition is called eigenvalue decomposition (Eigen Value Decomposition, EVD); more precisely, since matrix A is a symmetric matrix, this decomposition is a spectral decomposition.

Figure 6. Decompose $\mathbf{A}$ into “Rotation → Scaling → Rotation”

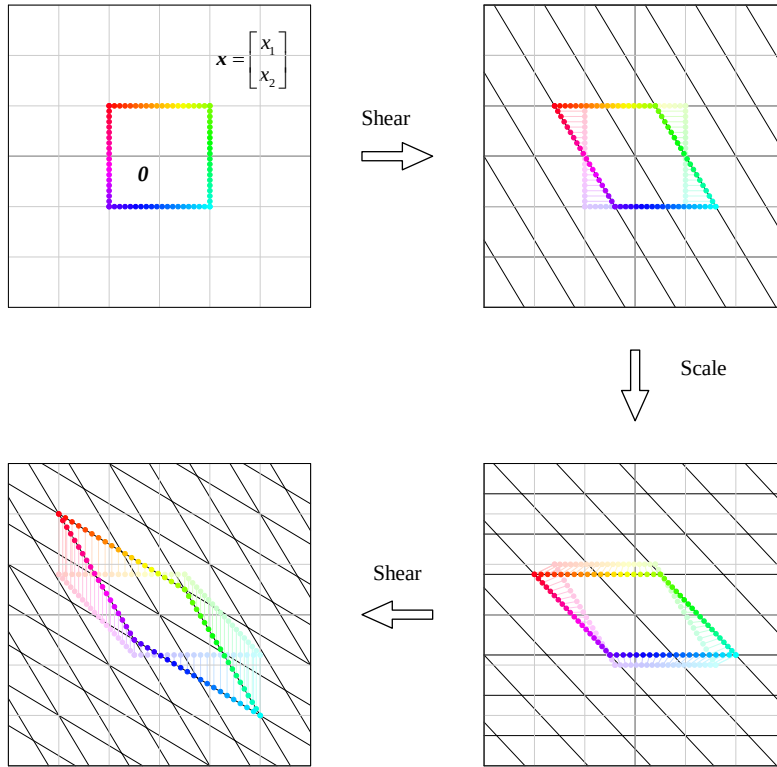
As another example, as shown in Figure 7, we can decompose matrix $\mathbf{A}$ into “shear → scaling → shear”, with the corresponding matrix multiplication being

$$\begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -0.6 & 1 \end{bmatrix} @ \begin{bmatrix} 1.25 & 0 \\ 0 & 0.8 \end{bmatrix} @ \begin{bmatrix} 1 & -0.6 \\ 0 & 1 \end{bmatrix}$$

Note that in Figure 7, one shear is along the horizontal axis and the other is along the vertical axis.

This decomposition is called LDL decomposition; similar decompositions include LU decomposition and Cholesky decomposition.

Different matrix decompositions correspond to different algorithms, each with its own geometric interpretation; later chapters of this book will introduce various common matrix decompositions.

Figure 7. Decompose A into “Shear $\rightarrow$ Scaling $\rightarrow$ Shear”

In General, $AB \neq BA$

As mentioned earlier in this book, even when the matrix products AB and BA both exist, in general

$$AB \neq BA$$

For example, AB and BA may result in matrices of different shapes.

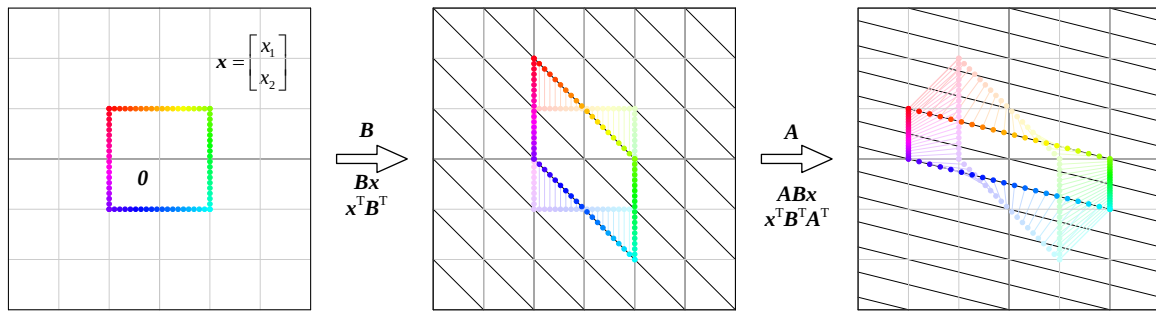
Even when AB and BA have the same shape, the geometric transformations they represent may still differ.

Given the following matrices A and B

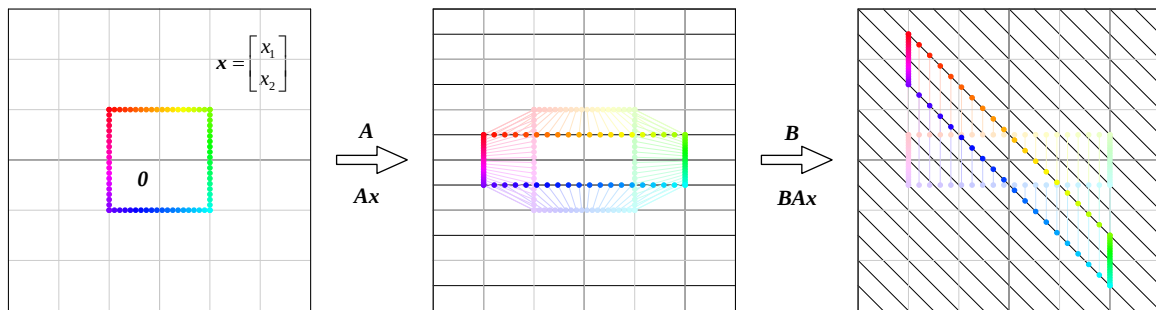
$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Matrix A corresponds to scaling, and B corresponds to shearing.

The matrix multiplication ABx represents first shearing x (B), then scaling (A) – shown specifically in Figure 8.

Figure 8. $ABx = y$'s Step-by-Step Geometric Operations

The matrix multiplication BAx represents first scaling x (A), then shearing (B) – shown specifically in Figure 9.

Figure 9. $BAx = y$'s Step-by-Step Geometric Operations

A Special Case: $AB = BA$

There are some special cases where the matrix product $AB = BA$. Let's discuss them below.

First, if A and B are both identity matrices of the same shape I , then clearly $AB = BA$. The identity matrix I means the geometric object undergoes no geometric change at all.

If 2×2 matrices A and B are both scaling matrices, for example

$$A = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 2/3 \end{bmatrix}$$

If you work it out, you'll find

$$AB = BA$$

As shown in Figure 10, even if we swap the order of the scaling matrices A and B , the final result is exactly the same.



Please compute the matrix products AB and BA .

This shows that, in this special case, matrix multiplication satisfies commutativity.

Let's look at another example.

Given 2×2 matrices A and B that are both shear matrices along the vertical axis, for example

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3/2 & 1 \end{bmatrix}$$

As shown in Figure 11, clearly $\mathbf{ABx}$ and $\mathbf{BAx}$ give exactly the same result.

? Please compute the matrix products $\mathbf{AB}$ and $\mathbf{BA}$.

Let's look at a third example.

Given 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ that are both rotation matrices about the origin, for example

$$\mathbf{A} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}$$

As shown in Figure 12, clearly $\mathbf{ABx}$ and $\mathbf{BAx}$ give exactly the same result.

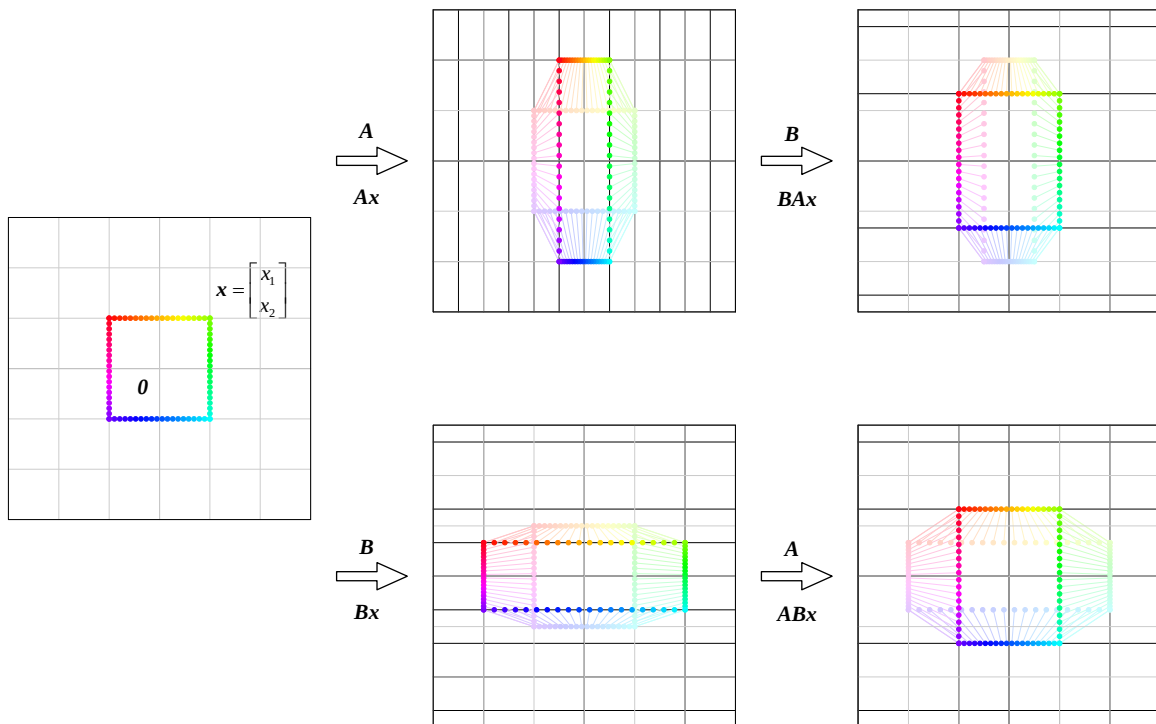
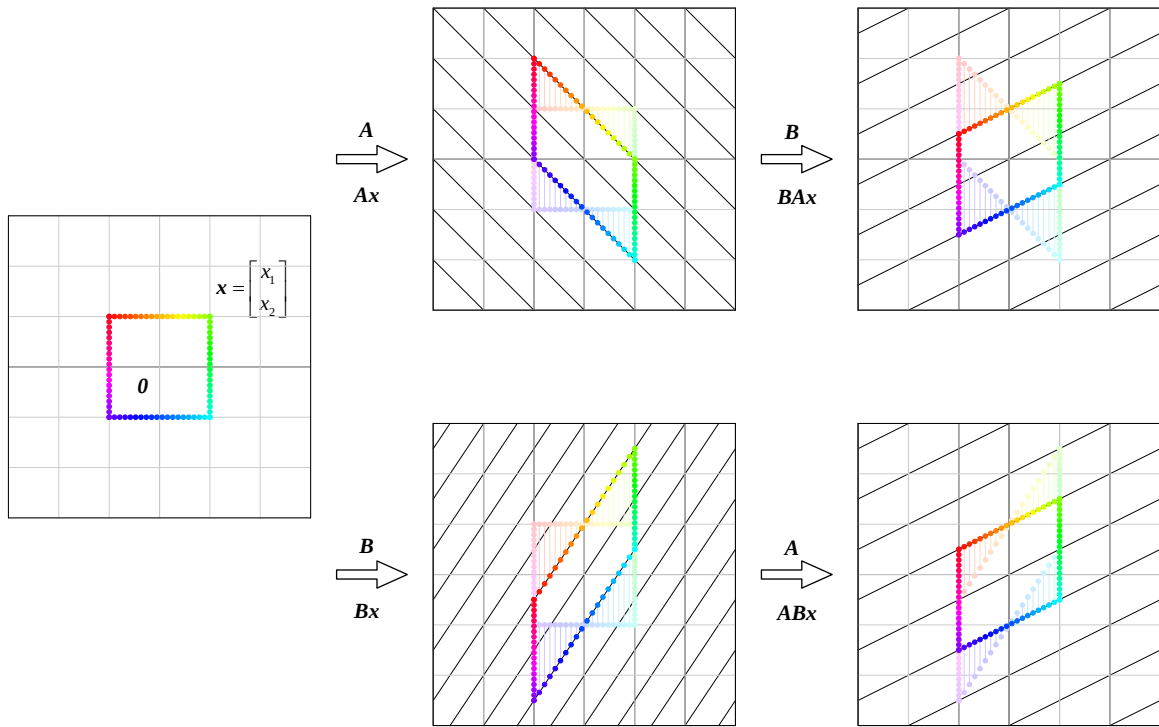
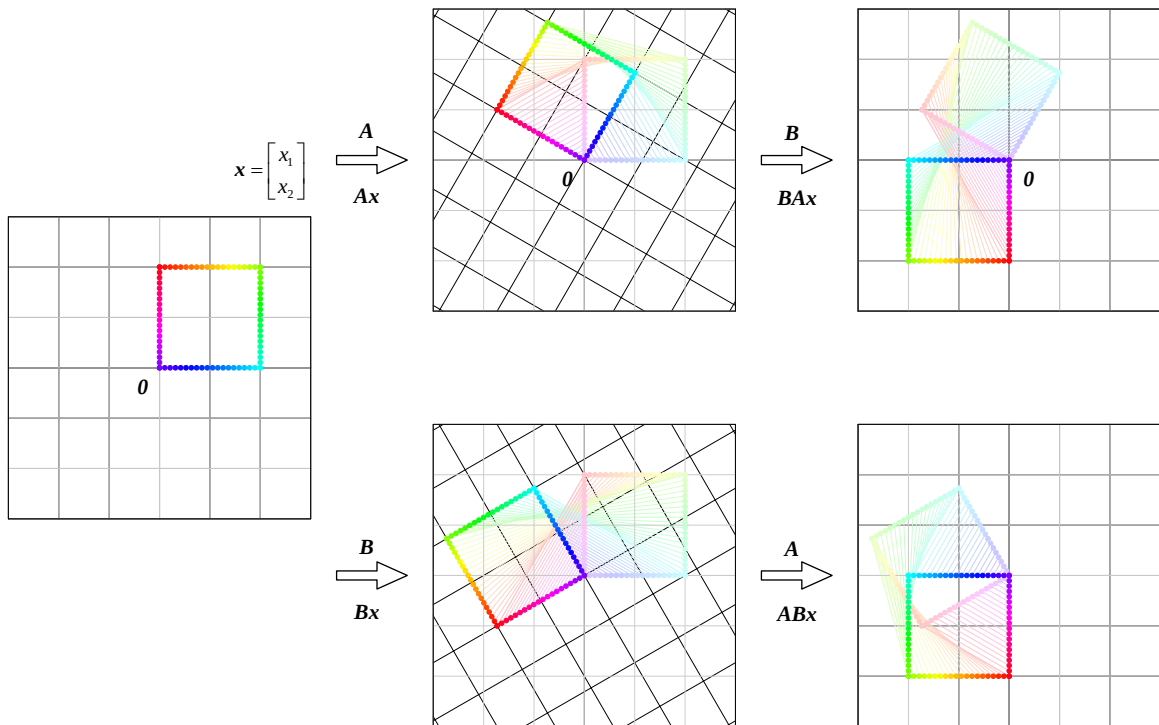


Figure 10. 2×2 Matrices $\mathbf{A}$ and $\mathbf{B}$ Are Both Scaling Matrices, $\mathbf{AB} = \mathbf{BA}$

Figure 11. 2×2 Matrices A and B Are Both Shear Matrices Along the Vertical Axis, $AB = BA$ Figure 12. 2×2 Matrices A and B Are Both Rotation Matrices About the Origin, $AB = BA$

Later chapters of this book will explain the special matrix-multiplication cases above from the perspective of these underlying geometric mechanisms.

Matrix Power

Matrix power (power of a matrix) refers to the repeated product of the square matrix $\mathbf{A}$ with itself,

$$\begin{aligned} \mathbf{A}^0 &= \mathbf{I} \\ \mathbf{A}^1 &= \mathbf{A} \\ \mathbf{A}^k &= \underbrace{\mathbf{A} @ \mathbf{A} @ \cdots @ \mathbf{A}}_{k \text{ times}} \end{aligned}$$

Note that matrix power requires the matrix to be square. If it isn't square, the matrix multiplication cannot be repeated, since the matrix dimensions wouldn't match.

Also, k in the equation above can be a negative integer, in which case matrix $\mathbf{A}$ is required to be invertible.

From a geometric perspective, a matrix power can be interpreted as the repeated application of the linear transformation $\mathbf{A}$,

$$\mathbf{A}^k \mathbf{x} = \mathbf{A}(\mathbf{A}^{k-1} \mathbf{x})$$

For example, given the matrix $\mathbf{A}$ below

$$\mathbf{A} = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$$

As shown in Figure 13, matrix $\mathbf{A}$ applied repeatedly to vector $\mathbf{x}$ ($\mathbf{A}^k \mathbf{x}$) enlarges the planar geometric shape while continuously rotating it.

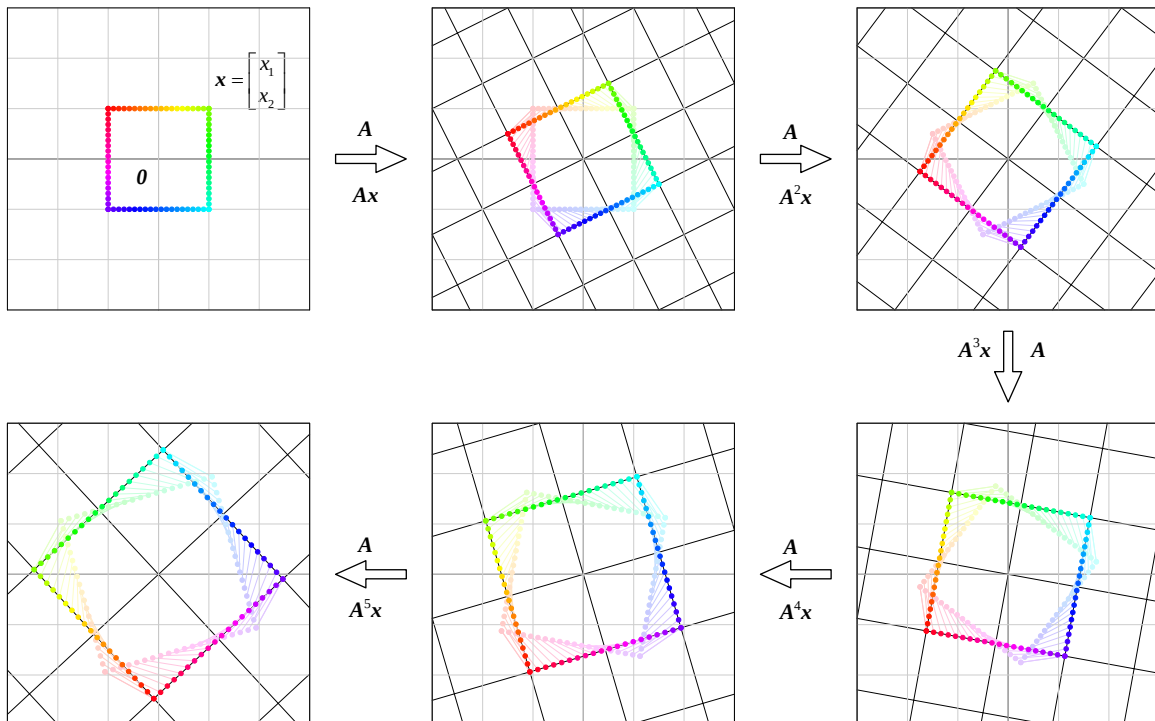


Figure 13. Powers of the 2×2 Matrix $\mathbf{A}$

$\mathbf{AB} = \mathbf{0}$

If the matrix product of A and B is the zero matrix, that is, $AB = O$, this doesn't mean A and B are zero matrices; nor does it mean $BA = O$.

For example, given the 2×2 matrices A and B

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

As shown in Figure 14, ABx causes the square to “collapse” to the origin.

However, the result of BA is not the zero matrix; as shown in Figure 15, the square undergoes a dimension reduction.

 Please compute the two matrix products AB and BA .

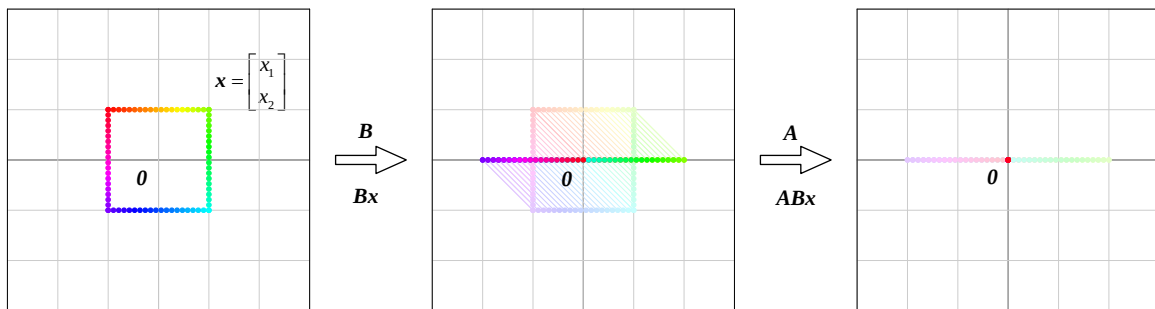


Figure 14. 2×2 Matrices A and B Have a Matrix Product of the Zero Matrix, $AB = O$

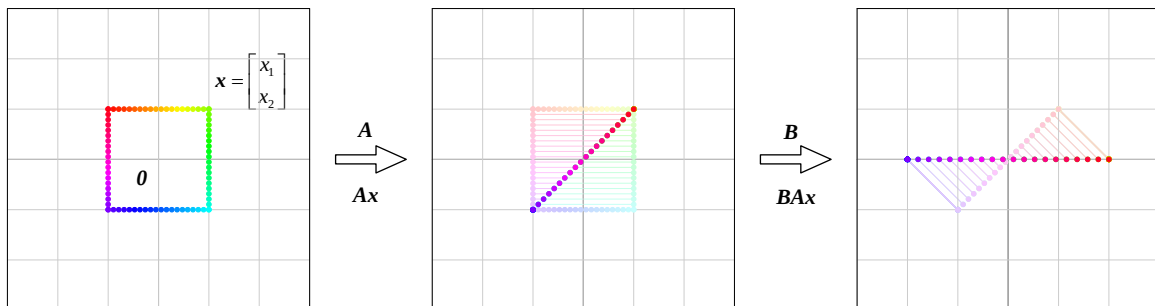


Figure 15. 2×2 Matrices A and B Have a Matrix Product of the Zero Matrix, but BA Is Not O

Also, if $A = B$, then $AC = BC$ or $CA = CB$.

Unless C is invertible, $AC = BC$ does not let us conclude that $A = B$.

Computational Complexity

For example, take the following product of 5 matrices

$$A_{m \times p_1} A_{p_1 \times p_2} A_{p_2 \times p_3} A_{p_3 \times p_4} A_{p_4 \times n}$$

As shown in the figure below, all the intermediate dimensions cancel out during the computation, leaving only the leading and trailing dimensions m and n .

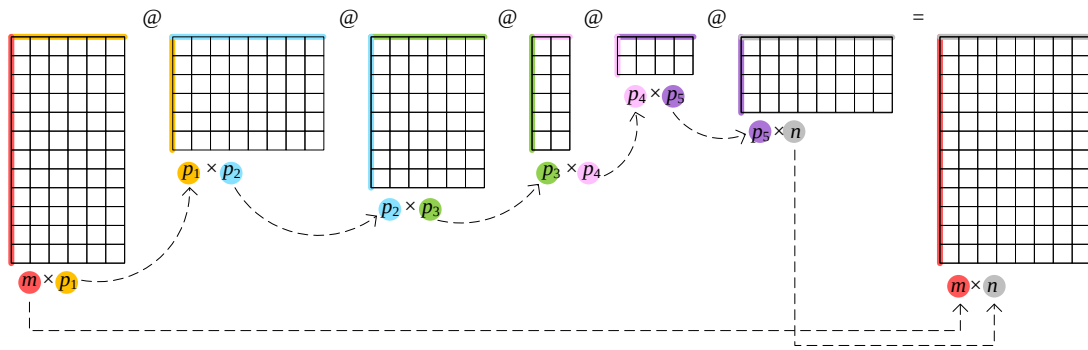


Figure 16. Consecutive Matrix Multiplication

In practice, however, the order of consecutive multiplication significantly affects computational efficiency, because the complexity of matrix multiplication depends on both the sizes of the matrices and the order of multiplication.

The complexity of matrix multiplication describes the number of computational steps needed to complete it. This complexity depends on the matrix dimensions as well as the number of additions and multiplications involved. Since the total computation is dominated mainly by multiplications, the complexity discussed in this section only considers the number of multiplications.

For example, for two matrices A ($m \times p$) and B ($p \times n$), in the matrix multiplication $C = AB$, each element in the result C is computed as the vector inner product of one row of A and one column of B . This means computing each element requires p multiplications and $p - 1$ additions.

The matrix multiplication $C = AB$ has $m \times n$ elements, and each element requires p multiplications. So the total complexity is $m \times n \times p$.

Now let's look at an example. As shown in the figure below, four matrices are multiplied in sequence: A (5×10), B (10×4), C (4×20), and D (20×5). Below, we compare two matrix-multiplication orders; please analyze the other possible orders and their complexities on your own.

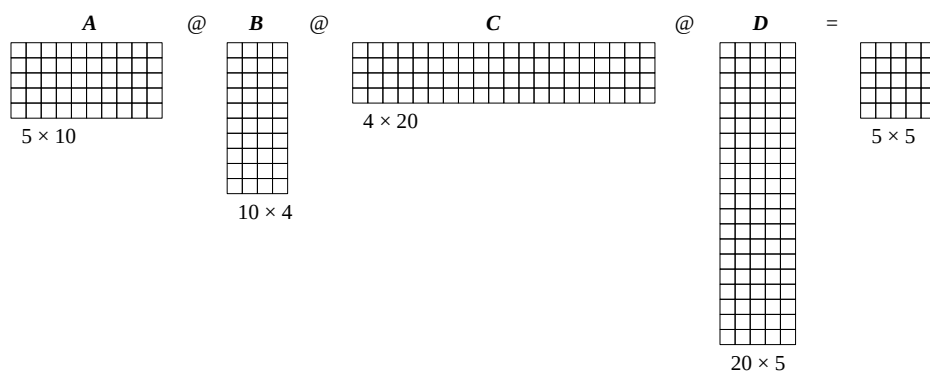


Figure 17. Optimizing the Order of Consecutive Matrix Multiplication

The first approach follows the order of $ABCD$ directly, that is, $((A @ B) @ C) @ D$, the second approach, based on the associativity of multiplication, first computes AB and CD separately, then multiplies the two results together, that is, $(A @ B) @ (C @ D)$.

The step-by-step complexity of the first consecutive-multiplication approach is:

- ▶ Step 1, $A @ B$ has complexity 200 ($5 \times 10 \times 4$).
- ▶ Step 2, $((A @ B) @ C)$ has complexity 400 ($5 \times 4 \times 20$); $(A @ B)$ has shape 5×4 .
- ▶ Step 3, $((A @ B) @ C) @ D$ has complexity 500 ($5 \times 20 \times 5$); $((A @ B) @ C)$ has shape 5×20 .

The total complexity of the first consecutive-multiplication approach is 1100 ($200 + 400 + 500$).

The step-by-step complexity of the second consecutive-multiplication approach is:

- ▶ Step 1, $A @ B$ has complexity 200 ($5 \times 10 \times 4$).
- ▶ Step 2, $C @ D$ has complexity 400 ($4 \times 20 \times 5$).
- ▶ Step 3, $(A @ B) @ (C @ D)$ has complexity 100 ($5 \times 4 \times 5$); $(A @ B)$ has shape 5×4 , $(C @ D)$ has shape 4×5 .

The total complexity of the second consecutive-multiplication approach is 700 ($200 + 400 + 100$).

This example shows that, in consecutive matrix multiplication, choosing the right computation order is crucial. Prioritizing the elimination of larger intermediate dimensions during computation can significantly reduce the subsequent workload, greatly lowering the overall complexity.

Optimizing the computation order isn't just a mathematical trick — it's also an important foundation for efficiently carrying out large-scale linear algebra tasks. This is especially true in machine learning and data science, where matrix operations are widely used for training models, processing high-dimensional data, and computing eigendecompositions. Optimizing the order of matrix multiplication can significantly improve computational performance, particularly when handling the forward and backward passes of neural networks.

Furthermore, modern optimization methods also make use of a matrix's structural characteristics. For example, sparse matrices can greatly reduce the number of nonzero elements involved, speeding up computation; block matrices can fully exploit parallel-computing resources in block-wise computation. These methods provide a diverse set of tools and ideas for further optimizing matrix operations, driving the broad application of linear algebra in computing. Optimizing the computation order, exploiting structural characteristics, and using advanced algorithms together form the theoretical and practical framework for efficient matrix operations.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Use a random-positive-integer generator to create four 2×2 matrices A , B , C , and D , then write your own Python code to compute the consecutive-multiplication result for every permutation of these matrices (for example $A @ B @ D @ C$).

Q2. Given the following matrix A that scales along the horizontal axis, and B that scales along the vertical axis, does $AB = BA$

▶ $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

Q3. Given the following matrix A that shears along the horizontal axis, and B that shears along the vertical axis, does $AB = BA$

▶ $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Q4. Given several shear matrices along the horizontal axis, please compute the matrix product $ABCD$ – what pattern do you notice?

▶ $A = \begin{bmatrix} 1 & k_1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & k_2 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & k_3 \\ 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & k_4 \\ 0 & 1 \end{bmatrix}$

Q5. Given the following different matrix A , please compute A^8 , and explain the effect of the linear transformation from a geometric perspective.

▶ $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$